SUBIN SIMON

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PROFESSIONAL SUMMARY

MTech Artificial Intelligence student at Amrita Vishwa Vidyapeetham with a published research paper (ICTIS 2026) on deep learning-based rainfall severity prediction using the Mamba/SSM architecture. Hands-on experience in computer vision, time-series forecasting (LSTM, Mamba-SSM), end-to-end ML model development and deployment on AWS. Proficient in Python, PyTorch, TensorFlow, and Scikit-learn. Seeking ML/AI Engineer roles where I can apply research-grade deep learning skills to real-world production systems.

EDUCATION

Master of Technology - Artificial Intelligence — 17/2024 – present
Amrita Vishwa Vidyapeetham,Coimbatore, India
CGPA: 7.2

Bachelor of Technology - Computer Science and Engineering — 06/2018 – 06/2022
Karunya Institute of Technology and Sciences, Coimbatore, India
CGPA : 8.52

SKILLS

Core ML & AI: Supervised Learning, Classification & Regression, Clustering, Model Evaluation, Feature Engineering, Data Preprocessing, Deep Learning, Computer Vision,Time Series Forecasting,LSTM,Mamba-SSM, Neural Networks, Large Language Models (LLMs)

Computer Vision: OpenCV, Image Classification, Data Augmentation, Image Preprocessing, Facial Emotion Recognition, Object Detection Basics

Libraries & Frameworks: TensorFlow,Keras,PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Cloud: Amazon Web Services(AWS)- EC2, S3, Lambda.

Tools & IDEs: Jupyter Notebook,Google Colab,VS Code, Git, Github

Programming Languages: Python, C / C++, MySQL,HTML/CSS

Operating Systems: Windows, Linux (Basic commands)

Soft Skills: Communication, Leadership, Team Collaboration, Active Listening, Willingness to Learn.

PROJECTS

Autoregressive Mamba Based Structured State Space Model for Regional Monsoon Rainfall Severity Prediction in Coimbatore
Tech Stack: Python · PyTorch · Mamba-SSM · ERA5-Land · Scikit-learn
- Developed a deep learning framework using **Mamba Structured State Space Models (SSM)** for regional monsoon rainfall severity prediction.
- Processed and aggregated ERA5-Land meteorological data from hourly to monthly resolution for temporal analysis.
- Engineered seasonal and temporal features to improve long-range dependency learning.
- Achieved 88.9% classification accuracy with a macro F1-score of 0.64 under class imbalance conditions.
- Implemented confusion matrix analysis and macro-metric evaluation for robust model validation.
- Compared model performance against baseline LSTM architectures for sequence forecasting.

Facial Emotion Recognition using Deep Convolutional Neural Networks
Tech Stack: Python, PyTorch/TensorFlow, OpenCV, CNN, FER-2013
- Built a **Convolutional Neural Network (CNN)** model for facial emotion recognition using the FER-2013 dataset.
- Performed image preprocessing, normalization, and data augmentation to improve model generalization.
- Classified emotional states including sadness, anger, disgust, and happiness for mental health assessment support.
- Improved detection robustness by eliminating manual feature engineering techniques.
- Evaluated model performance using accuracy and validation loss metrics.

Student Mark Prediction Using Machine Learning Techniques
Tech stack: Python.Scikit-learn, Pandas, Flask.

- Developed machine learning models to predict student academic performance using demographic and behavioral features.
- Conducted exploratory data analysis (EDA), feature preprocessing, and model evaluation.
- Compared multiple regression and classification algorithms using Scikit-learn.
- Deployed the trained ML model as a Flask web application for real-time prediction.

Infinity - An NFT Marketplace On Ethereum
Tech stack: Ethereum

- Developed a decentralized NFT marketplace allowing users to list,buy and sell Digital Assets.
- Implemented secure ownership transfer through smart contracts, handling transactions between buyers and sellers.
- Enabled revenue generation via listing fees, contributing to sustainable marketplace operations.
- Ensured seamless asset listings and transaction process using ethereum blockchain.

PUBLICATIONS

Autoregressive Mamba Based Structured State Space Model for Regional Monsoon Rainfall Severity Prediction in Coimbatore — 04/2026
Published at ICTIS Thailand

- Proposed a Mamba-based deep learning architecture for rainfall severity prediction using ERA5-Land climate datasets.
- Achieved 88.89% prediction accuracy and improved long-range temporal dependency modeling over baseline LSTM models.
- Implemented structured state-space sequence modeling using PyTorch.

CERTIFICATES

- Introduction to Large Language Models (NPTEL)
- Responsible & Safe AI Systems (NPTEL)
- Applied Social Network Analysis in Python (NetworkX , Network Prediction)
- India_Internship_Karunya University_Introduction to Cybersecurity
- Cambridge English Level1, ESOL International (Business Vantage)

LANGUAGES

- English (Professional)
- · Hindi (Conversational)
- Malayalam (Native)